

BINDU CHANDRA SHEKAR REDDY

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Professional Summary

Full-stack web developer with one year of experience in HTML, CSS, JavaScript, PHP, and Python. Skilled in building responsive web applications, real-time systems, and RESTful APIs. Experienced with SQL databases, version control, and debugging. Quick to learn, adaptable, and a proactive team player focused on delivering efficient, user-friendly solutions and contributing to project success.

Skills

- **Programming:** PHP, Python, Java, C, SQL, JavaScript
- **Tools & Platforms:** Docker, Kubernetes, Git, VS Code, Xcode, Google Kubernetes Engine (GKE), Google Colab

Experience

Teleindia Networks Pvt. Ltd

Web Developer

Bengaluru, India · Jun 2023 - Jul 2024

- Developed and maintained responsive web applications using React, PHP, and Node.js, applying CSS modules and Jest testing, which improved usability and cross-device performance as reflected in positive user feedback
- Implemented RESTful APIs and optimized MySQL queries, reducing average response time by 30%.
- Collaborated with designers and product teams, using React and PHP, to deliver full-stack solutions that met business requirements and enabled on-time release of key features
- Integrated Java-based backend modules with Spring Boot, ensuring seamless data flow between client and server layers, which eliminated data synchronization errors
- Participated in incident management, debugging and resolving production issues to enhance reliability by 20%.

California State University Sacramento

Teaching Assistant

Feb 2024 - Dec 2024

- Assisted students in learning Java fundamentals, focusing on object-oriented design, debugging, and coding best practices, which helped them write clearer and more reliable code.
- Conducted lab sessions and gave targeted feedback on assignments through collaborative tools such as Google Colab and Git, which strengthened students' programming skills and deepened their conceptual understanding

Instructional Student Assistant

Aug 2025 - Present

- Guided students in applying object-oriented programming and computer-graphics concepts using Java, helping them complete weekly coding assignments and deepening their ability to create interactive visualizations
- Supported project reviews and lab exercises to enhance understanding of animation, event handling, and software design.

Nanorobotics Embed Technologies

Machine Learning Intern

Bengaluru, India · Sep 2021 - Oct 2021

- Developed and tested ANN models in Python with NumPy, Pandas, and Matplotlib, achieving reliable predictions that supported downstream embedded analytics
- Programmed Python-based software modules for embedded devices and built ML inference pipelines, enabling real-time analytics for improved device performance.
- Presented findings on ML integration for embedded robotics to the engineering team, informing design decisions and advancing the project roadmap

Projects

NeuroFusionGPT – Multimodal Stress Detection System

- Built a **multimodal deep learning model** that predicts **5 stress levels** by fusing **EEG (brain signals)** and **ECG (heart signals)** using a feature-level neural fusion architecture.
- Implemented **Transformer-based EEG modeling (16 features)** and **MLP-based ECG analysis (188 features)** to learn physiological stress patterns from **95K+ signal samples**.
- Developed an **end-to-end ML pipeline** including preprocessing, feature scaling, multimodal fusion, inference, and evaluation using **accuracy, precision, recall, F1-score, and confusion matrix**.
- Integrated an **AI wellness feedback module** that generates personalized stress-management guidance based on predicted stress levels; trained models in **Google Colab (GPU)** and deployed locally in Python.

ML-based Medical Diagnostics for Infectious Diseases

- Selected for presentation at IEEE SysCon (Systems Conference) for scalability and real-world diagnostic relevance.
- A supervised ML system to predict infection status from epidemiological surveillance data in high-noise, resource-limited settings.
- Engineered novel clinical features such as comorbidity counts and symptom-to-admission intervals to strengthen predictive signals from demographic data.
- Optimized an XGBoost classifier to achieve 0.80 ROC-AUC and 0.73 recall, prioritizing reduced false negatives for public health safety.

IoT-Based Health Monitoring System with Edge Computing

- Developed an iOS frontend for real-time health tracking via connected IoT devices.
- Integrated edge computing for local data processing to reduce latency and improve performance.
- Containerized services using Docker and deployed on Google Kubernetes Engine (GKE) for scalable backend support.

Plant Leaf Disease Detection Using CNN

- Engineered a CNN model using transfer learning (VGG16, ResNet50), achieving ~95% classification accuracy on plant disease datasets.
- Processed and visualized data using Python libraries (NumPy, Pandas, Matplotlib).
- Delivered a real-time classification system enabling early disease detection and preventive action for farmers.

Education

California State University, Sacramento

Master of Science, Computer Science

Aug 2024 - May 2026

Sacramento, CA

KNS Institute of Technology (VTU)

Bachelor of Engineering, Computer Science and Engineering

Aug 2018 - Jun 2022

Bengaluru, India